

Which of the following thermodynamic parameters are state functions?

I. Enthalpy

II. Temperature

III. Heat

☐ A. II only [11%]

✓ ☒ B. I and II only [57%]

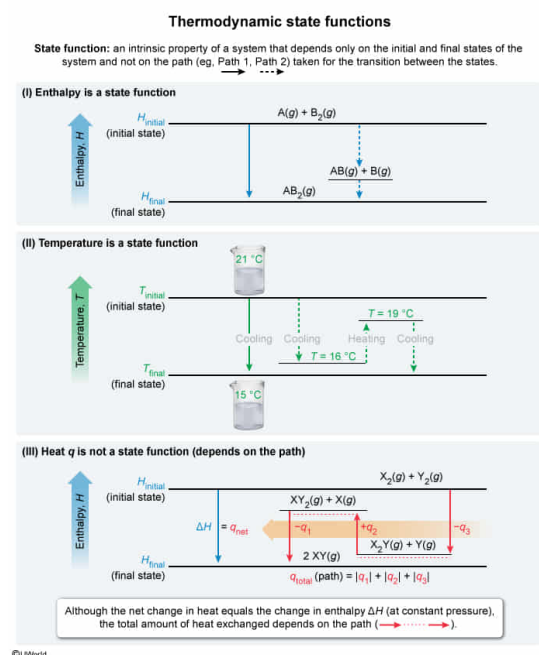
☐ C. II and III only [11%]

☐ D. I, II, and III [19%]

Correct

57% answered correctly

Explanation:



A **state function** describes an intrinsic property of a thermodynamic system whose value depends only on the **state** of the system. As such, a state function is **independent of the path taken** (ie, the process by which the state was achieved) to reach a specific value.

(Number I) For a thermodynamic system held at constant pressure, the change in total energy is described by the change in enthalpy ΔH and is numerically equivalent to the *net amount* of heat energy transferred in or out of a system. The value of ΔH depends only on the initial state (H_{initial}) and the final state (H_{final}) of the system, and therefore is a state function.

(Number II) Temperature measures the **average kinetic energy** of a system (ie, a property of the state). Accordingly, the change in temperature ΔT of the system depends only on the initial and final temperatures, irrespective of the path taken. Therefore, temperature is a state function.

(Number III) Heat is defined as the transfer of energy between systems of different temperatures, and is not an intrinsic property of a system. The *total amount* of heat exchanged with a system is dependent on the path taken to achieve the final state. Therefore, heat is a path function, not a state function.

Educational objective:

A state function depends only on the state of a system, irrespective of the path taken to achieve a given state.

Time spent:0 sec

Subject:
General Chemistry

QID:403174

System:
Thermodynamics, Kinetics & Gas Laws